

Section A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1.				8	1			1		
2.				Group 2			1	1		
3.				neon core 3s 1↓ 3p 1↓ 1↓ 1↓ 3d 1 1 1 1 1 4s 1	1			1		
4.	(a)			$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$		1		1		
	(b)			$M_r \text{H}_3\text{PO}_4 = 98(.03)$ and either $M_r \text{HCl} = 36.5(1)$ or $M_r \text{PCl}_5 = 208.5$ and $M_r \text{H}_2\text{O} = 18(.02)$ (1) atom economy = 35 % (1) ecf possible from part (a)		2		2	1	
5.	(a)			ability to attract electrons in a covalent bond / a shared electron pair	1			1		
	(b)			increase in number of protons / charge on the nucleus	1			1		
6.	(a)			$\text{Cl}_2 + 2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{Cl}^-$	1			1		
	(b)			silver nitrate solution / $\text{AgNO}_3(\text{aq})$ gives cream/off-white precipitate	1			1		1
				Section A total	6	3	1	10	1	1